Q1.

The discrete random variable X has probability distribution defined by

$$P(X = x) = \begin{cases} 0.1 & x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of $P(4 \le X \le 7)$

Circle your answer.

0.2 0.3 0.4 0.5

(Total 1 mark)

Q2.

The discrete random variable R has the following probability distribution.

r	–2	0	a	4
P(R = r)	0.3	b	c	0.1

It is known that E(R) = 0.2 and Var(R) = 3.56

Find the values of a, b and c.

(Total 4 marks)

Q3.

In a computer game, players try to collect five treasures. The number of treasures that Isaac collects in one play of the game is represented by the discrete random variable X.

The probability distribution of X is defined by

$$P(X = x) = \begin{cases} \dfrac{1}{x+2} & x = 1, 2, 3, 4 \\ k & x = 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) (i) Show that $k = \dfrac{1}{20}$.

(2)

(ii) Calculate the value of E(X).

(2)

(iii) Show that Var(X) = 1.5275.

(3)

(iv) Find the probability that Isaac collects more than 2 treasures.

(2)

(b) The number of points that Isaac scores for collecting treasures is Y where

$$Y = 100X - 50$$

Calculate the mean and the standard deviation of Y.

(4)

(Total 13 marks)

Q4.

(a) Joshua plays a game in which he repeatedly tosses an unbiased coin. His game concludes when he obtains either a head or 5 tails in succession.

The random variable N denotes the number of tosses of his coin required to conclude a game.

By completing **Table 1** below, calculate $\mathrm{E}(N)$.

Table 1

n	1	2	3	4	5
P(N = n)			$\frac{1}{8}$		$\frac{1}{16}$

(4)

(b) Joshua's sister, Ruth, plays a separate game in which she repeatedly tosses a coin that is **biased** in such a way that the probability of a head in a single toss of her coin is $\frac{1}{4}$. Her game also concludes when she obtains either a head or 5 tails in succession.

The random variable M denotes the number of tosses of her coin required to conclude her game.

Complete **Table 2** below.

Table 2

m	1	2	3	4	5
P(M = m)	$\frac{1}{4}$	$\frac{3}{16}$			

(3)

(c) (i) Joshua and Ruth play their games simultaneously. Calculate the probability that Joshua and Ruth will conclude their games in an equal number of tosses of their coins.

(5)

(ii) Joshua and Ruth play their games simultaneously on 3 occasions. Calculate the probability that, on at least 2 of these occasions, their games will be concluded in an equal number of tosses of their coins. Give your answer to three decimal places.

(4)

(Total 16 marks)

Q5.

A house has a total of five bedrooms, at least one of which is always rented.

The probability distribution for R, the number of bedrooms that are rented at any given time, is given by

$$P(R = r) = \begin{cases} 0.5 & r = 1 \\ 0.4(0.6)^{r-1} & r = 2,\ 3,\ 4 \\ 0.0296 & r = 5 \end{cases}$$

(a) Complete the table below.

r	1	2	3	4	5
P(R = r)	0.5				0.0296

(2)

(b) Find the probability that fewer than 3 bedrooms are **not** rented at any given time.

(3)

(c) (i) Find the value of $E(R)$.

(2)

(ii) Show that $E(R^2) = 4.8784$ and hence find the value of Var(R).

(3)

(d) Bedrooms are rented on a monthly basis.

The monthly income, £M, from renting bedrooms in the house may be modelled by

$$M = 1250R - 282$$

Find the mean and the standard deviation of M.

(3)

(Total 13 marks)

Q6.

(a) A red biased tetrahedral die is rolled. The number, X, on the face on which it lands has the probability distribution given by

x	1	2	3	4
P($X = x$)	0.2	0.1	0.4	0.3

(i) Calculate E(X) and Var(X).

(3)

(ii) The red die is now rolled **three** times. The random variable S is the sum of the three numbers obtained.

Find E(S) and Var(S).

(2)

(b) A blue biased tetrahedral die is rolled. The number, Y, on the face on which it lands has the probability distribution given by

$$P(Y = y) = \begin{cases} \dfrac{y}{20} & y = 1, 2 \text{ and } 3 \\[2ex] \dfrac{7}{10} & y = 4 \end{cases}$$

The random variable T is the value obtained when the number on the face on which it lands is multiplied by 3.

Calculate E(T) and Var(T).

(6)

(c) Calculate:

(i) P($X > 1$);

(1)

(ii) P($X + T \leq 9$ **and** $X > 1$);

(4)

(iii) P($X + T \leq 9 \mid X > 1$).

(2)

(Total 18 marks)

Q7.

(a) Ali has a bag of 10 balls, of which 5 are red and 5 are blue. He asks Ben to select 5 of these balls from the bag at random.

The probability distribution of X, the number of red balls that Ben selects, is given in the **Table 1**.

Table 1

x	0	1	2	3	4	5

P $(X = x)$	$\frac{1}{252}$	$\frac{25}{252}$	$\frac{100}{252}$	a	$\frac{25}{252}$	$\frac{1}{252}$

(i) State the value of a.

(1)

(ii) Hence write down the value of E(X).

(1)

(iii) Determine the standard deviation of X.

(5)

(b) Ali decides to play a game with Joanne using the same 10 balls. Joanne is asked to select 2 balls from the bag at random.

Ali agrees to pay Joanne 90p if the two balls that she selects are the same colour, but nothing if they are different colours. Joanne pays 50p to play the game.

The probability distribution of Y, the number of red balls that Joanne selects, is given in **Table 2**.

Table 2

y	0	1	2
P($Y = y$)	$\frac{2}{9}$	$\frac{5}{9}$	$\frac{2}{9}$

(i) Determine whether Joanne can expect to make a profit or a loss from playing the game once.

(ii) Hence calculate the expected size of this profit or loss after Joanne has played the game 100 times.

(3)
(Total 10 marks)

Q8.

(a) The number of strokes, R, taken by the members of Duffers Golf Club to complete the first hole may be modelled by the following discrete probability distribution.

r	≤ 2	3	4	5	6	7	8	≥ 9
P($R = r$)	0	0.1	0.2	0.3	0.25	0.1	0.05	0

(i) Determine the probability that a member, selected at random, takes at least 5 strokes to complete the first hole.

(1)

(ii) Calculate E(R).

(2)

(iii) Show that Var(R) = 1.66.

(4)

(b) The number of strokes, S, taken by the members of Duffers Golf Club to complete the second hole may be modelled by the following discrete probability distribution.

s	≤ 2	3	4	5	6	7	8	≥ 9
P ($S = s$)	0	0.15	0.4	0.3	0.1	0.03	0.02	0

Assuming that R and S are independent:

(i) show that P($R + S \le 8$) = 0.24;

(5)

(ii) calculate the probability that, when 5 members are selected at random, at least 4 of them complete the first two holes in fewer than 9 strokes;

(3)

(iii) calculate P($R = 4 \mid R + S \le 8$).

(3)
(Total 18 marks)

Q9.

A small supermarket has a total of four checkouts, at least one of which is always staffed. The probability distribution for R, the number of checkouts that are staffed at any given time, is

$$P(R = r) = \begin{cases} \frac{2}{3}\left(\frac{1}{3}\right)^{r-1} & r = 1, 2, 3 \\ k & r = 4 \end{cases}$$

(a) Show that $k = \frac{1}{27}$.

(2)

(b) Find the probability that, at any given time, there will be at least 3 checkouts that are staffed.

(1)

(c) It is suggested that the total number of customers, C, that can be served at the checkouts per hour may be modelled by

$$C = 27R + 5$$

Find:

(i) E(C);

(3)

(ii) the standard deviation of C.

(4)
(Total 10 marks)

Q10.

Joanne has 10 identically-shaped discs, of which 1 is blue, 2 are green, 3 are yellow and 4 are red. She places the 10 discs in a bag and asks her friend David to play a game by selecting, at random and without replacement, two discs from the bag.

(a) Show that:

(i) the probability that the two discs selected are the same colour is $\frac{2}{9}$;

(2)

(ii) the probability that exactly one of the two discs selected is blue is $\frac{1}{5}$.

(2)

(b) Using the discs, Joanne plays the game with David, under the following conditions:

If the two discs selected by David are the same colour, she will pay him 135p. If exactly one of the two discs selected by David is blue, she will pay him 145p. Otherwise David will pay Joanne 45p.

(i) When a game is played, X is the amount, in pence, **won by David**. Construct the probability distribution for X, in the form of a table.

(2)

(ii) Show that $E(X) = 33$.

(2)

(c) Joanne modifies the game so that the amount per game, Y pence, that **she wins** may be modelled by

$$Y = 104 - 3X$$

(i) Determine how much Joanne would expect to win if the game is played 100 times.

(3)

(ii) Calculate the standard deviation of Y, giving your answer to the nearest 1p.

(4)
(Total 15 marks)

Q11.

A discrete random variable X has the probability distribution

$$\mathrm{P}(X = x) = \begin{cases} \dfrac{x}{20} & x = 1, 2, 3, 4, 5 \\ \dfrac{x}{24} & x = 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) Calculate $\mathrm{P}(X \geq 5)$.

(2)

(b) (i) Show that $\mathrm{E}\left(\dfrac{1}{X}\right) = \dfrac{7}{24}$.

(2)

(ii) Hence, or otherwise, show that $\mathrm{Var}\left(\dfrac{1}{X}\right) = 0.036$, correct to three decimal places.

(3)

(c) Calculate the mean and the variance of A, the area of rectangles having sides of length $X + 3$ and $\dfrac{1}{X}$.

(5)

(Total 12 marks)

Q12.

The number of mistakes, X, that Holli makes as a learner driver when she drives from Ampthill to Bedford can be modelled by the following discrete probability distribution:

x	≤ 1	2	3	4	5	6	≥ 7
P($X = x$)	0	0.40	0.25	0.18	0.12	k	0

(a) Find the value of k.

(1)

(b) Find:

(i) $\mathrm{E}(X)$;

(1)

(ii) $\mathrm{Var}(X)$.

(3)

(c) When Holli makes the return journey by the same route, the number of mistakes, Y, that she makes can be approximated by

$$Y = 2X - 3$$

Find:

(i) E(Y);

(1)

(ii) the standard deviation of Y.

(3)
(Total 9 marks)

Q13.

The number of fish, X, caught by Pearl when she goes fishing can be modelled by the following discrete probability distribution:

x	1	2	3	4	5	6	≥ 7
P($X = x$)	0.01	0.05	0.14	0.30	k	0.12	0

(a) Find the value of k.

(1)

(b) Find:

(i) E(X);

(1)

(ii) Var(X).

(3)

(c) When Pearl sells her fish, she earns a profit, in pounds, given by

$$Y = 5X + 2$$

Find:

(i) E(Y);

(1)

(ii) the standard deviation of Y.

(3)
(Total 9 marks)

Q14.

A discrete random variable X has probability distribution as given in the table.

x	1	2	3	4
P($X = x$)	p	p	p	$1 - 3p$

(a) Show that, for this to be a valid distribution, $0 \le p \le \frac{1}{3}$.

(3)

(b) (i) Find an expression, in terms of p, for E(X).

(1)

(ii) Show that Var(X) = $2p(7 – 18p)$.

(3)

(c) (i) Find the value of p for which Var(X) is a maximum.

(2)

(ii) Find the maximum value of the standard deviation of X.

(3)

(Total 12 marks)

Q15.

On a multiple choice examination paper, each question has five alternative answers given, only one of which is correct. For each question, candidates gain 4 marks for a correct answer but lose 1 mark for an incorrect answer.

(a) James guesses the answer to each question.

(i) Copy and complete the following table for the probability distribution of X, the number of marks obtained by James for each question.

x	4	–1
P($X = x$)		

(1)

(ii) Hence find E(X).

(2)

(b) Karen is able to eliminate two of the incorrect answers from the five alternative answers given for each question before guessing the answer from those remaining.

Given that the examination paper contains 24 questions, calculate Karen’s expected total mark.

(4)

(Total 7 marks)

Q16.

The number of aces, X, served by a tennis player during a set can be modelled by the following probability distribution.

x	0	1	2	3	4	5

P(X = x)	0.01	0.09	0.13	0.50	0.15	0.12

(a) Calculate the mean, μ, and the standard deviation, σ, of X.

(4)

(b) For a set selected at random, calculate P(μ – σ < X < μ + σ).

(3)
(Total 7 marks)

Q17.

A biased six-sided die is to be rolled once. The probabilities of scoring 1, 2, 3 and 4 are each equal to $\frac{1}{8}$. The probability of scoring 5 is equal to the probability of scoring 6. The score is represented by a random variable X.

(a) Copy and complete the following table to show the distribution fully.

x	1	2	3	4	5	6
P(X = x)	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$		

(1)

(b) Calculate the mean and variance of X.

(5)
(Total 6 marks)

Q18.

The probability distribution of a random variable X is given in the table below.

x	1	2	3
P(X = x)	$\frac{1}{2}$	$\frac{1}{3}$	p

(a) Find the value of p.

(1)

(b) Show that the mean of X is $\frac{5}{3}$ and find the variance of X.

(4)
(Total 5 marks)

Q19.

A village inn offers bed and breakfast and has four bedrooms available for customers. The number of bedrooms occupied each night may be modelled by the random variable, X, with the following probability distribution.

x	$P(X = x)$
0	0.22
1	0.25
2	0.20
3	0.14
4	0.19

(a) Calculate:

(i) the mean of X;

(ii) $E(X^2)$;

(iii) the standard deviation of X.

(6)

(b) Find the probability that the number of occupied bedrooms:

(i) is less than 3;

(ii) exceeds the mean;

(iii) exceeds the median.

(5)

(Total 11 marks)